

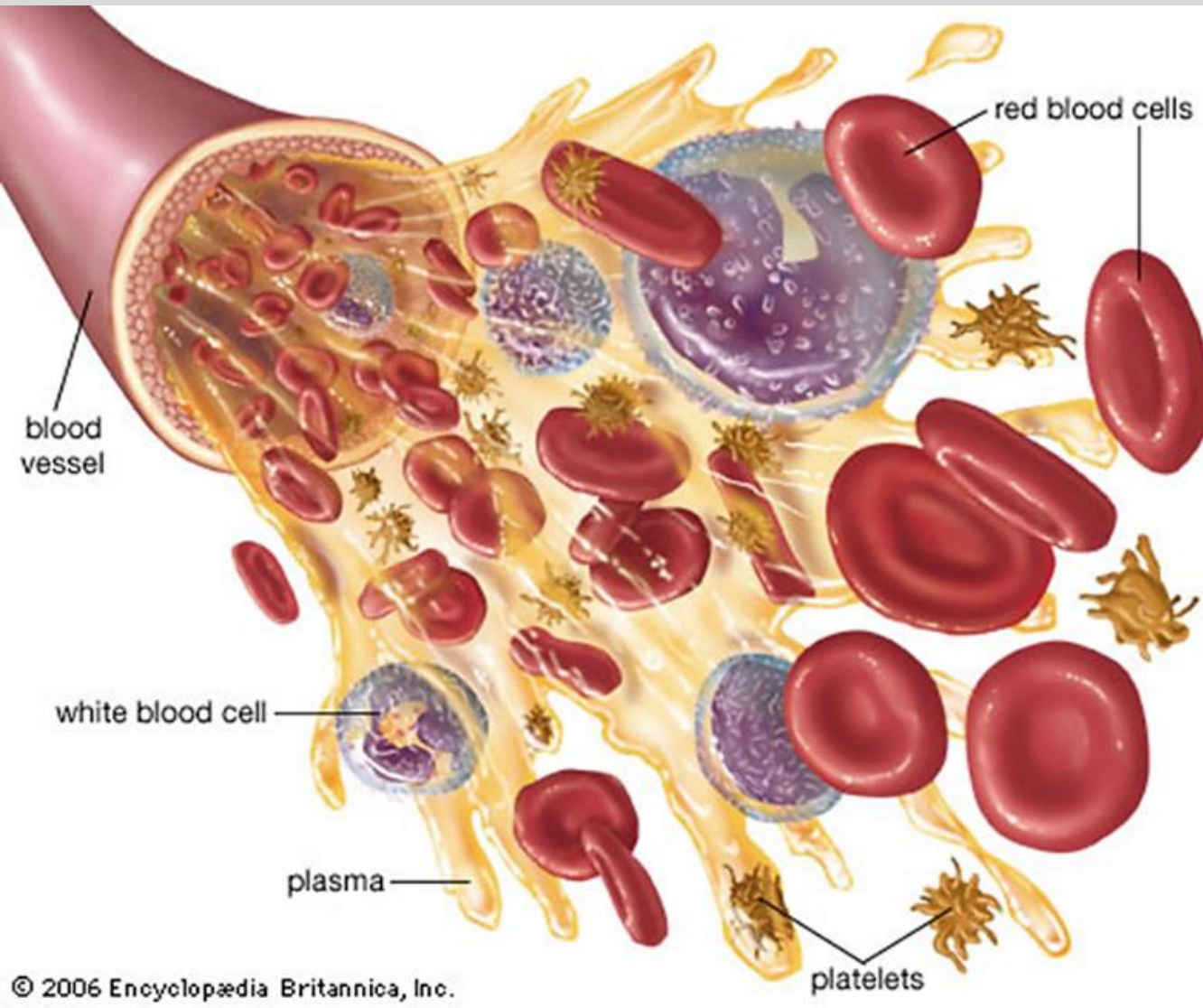
Session Objectives

1. Understand the **immunologic basis** of transfusion medicine, in particular the **ABO** and **Rh** blood group systems
2. Learn about the different **components** of whole blood and the **indication** for transfusing packed red blood cells, platelets, plasma, cryoprecipitate, albumin, IVIG, and factor concentrates
3. Understand and be able to apply the criteria for **appropriate** transfusion practice
4. Know the **alternatives** to transfusions
5. Understand the different types of **transfusion reactions**, their etiology and the evaluation and management of a patient who has experienced a transfusion reaction
6. Know the common **transfusion transmitted diseases** and how donors are screened for these

Blood

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~60% **plasma** (antigens and antibodies)

~40% formed elements (ex **RBCs**, **WBCs**) - antigens

Safe transfusion of blood

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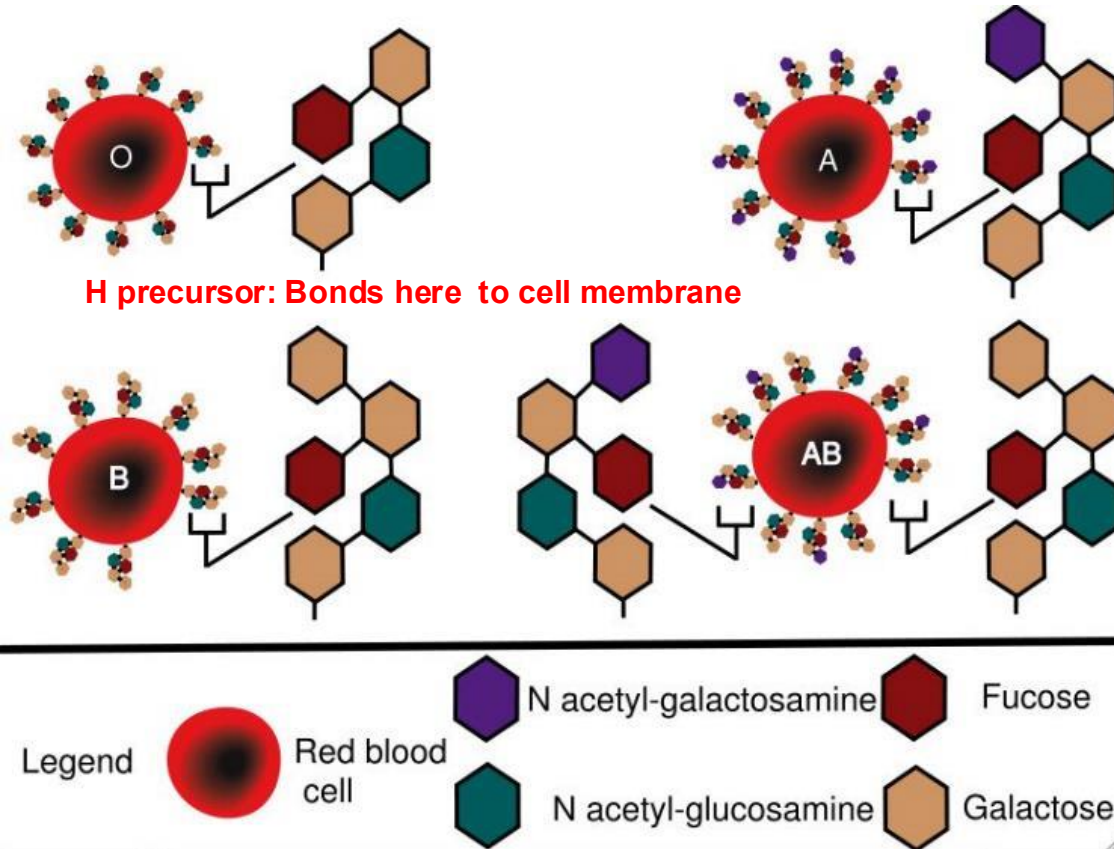
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- Humans belong to different **blood component groups** – (ex. RBC) surface antigens
 - With blood transfusion, **immune system** reacts with **antigens different from self** → antibodies against antigens they lack
 - For safe transfusions, ensure transfused antigens **match** patient's **blood type**
 - Potential complications/transfusion reactions include life threatening
 - Hemolysis → shock, acute renal failure, DIC
 - Anaphylaxis → from urticaria to shock

Properties of the major classes of antibodies in humans

PROPERTIES	CLASS OF ANTIBODY				
	IgM	IgD	IgG	IgA	IgE
Heavy chains	μ	δ	γ	α	ϵ
Light chains	κ or λ	κ or λ	κ or λ	κ or λ	κ or λ
Number of four-chain units	5	1	1	1 or 2	1
Percentage of total Ig in blood	10	<1	75	15	<1
Activates complement	++++	-	++	-	-
Crosses placenta	-	-	+	-	-
Binds to macrophages and neutrophils	-	-	+	-	-
Binds to mast cells and basophils	-	-	-	-	+

ABO blood group system **antigens**



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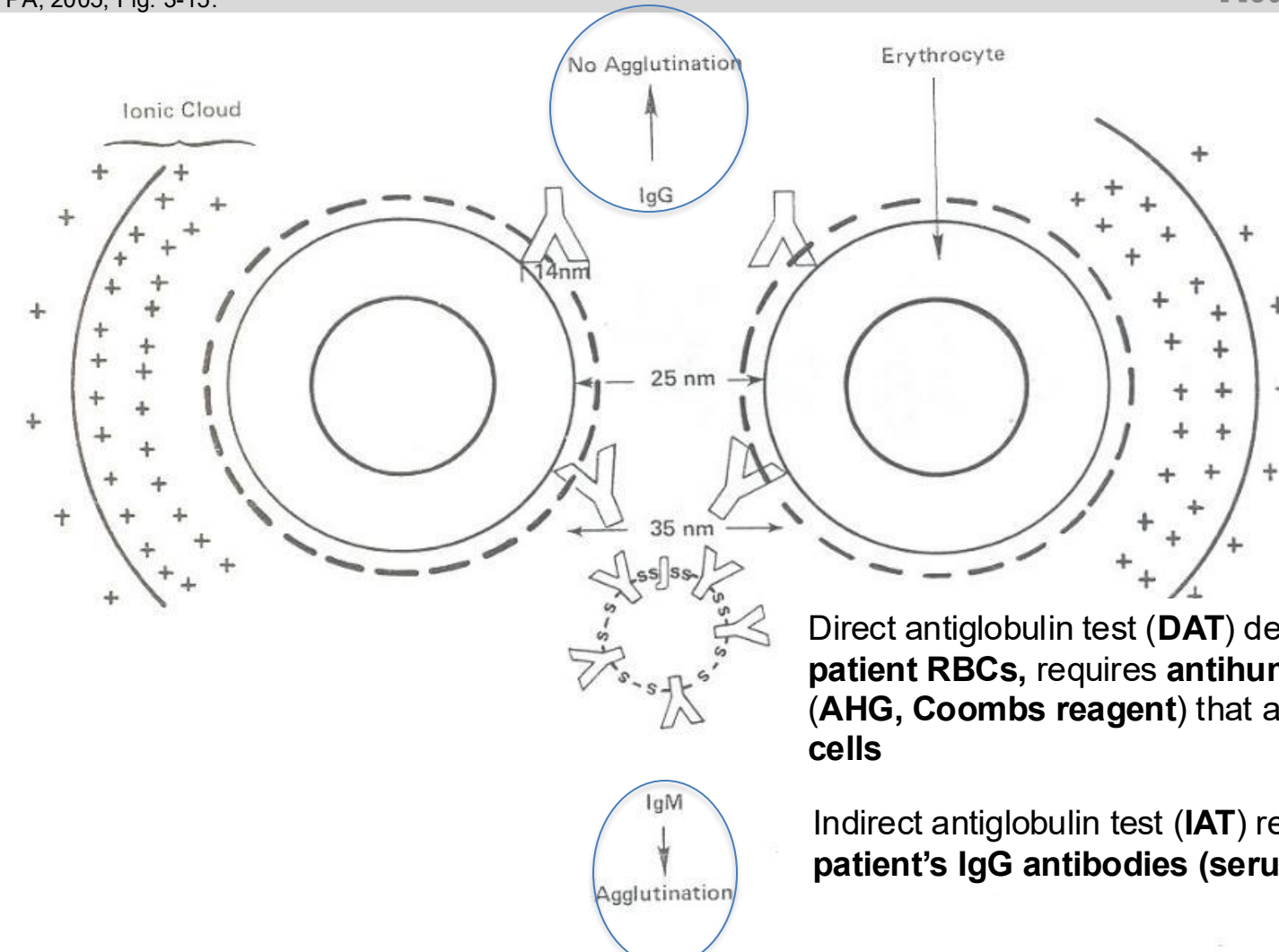
- ABO **antigens** are **most immunogenic** of all blood group antigens
- ABO genes produce enzymes that form **complex carbohydrates**
- Antigens vary due to terminal sugar epitopes
 - O (H): fucose
 - A: N-acetyl-galactosamine
 - B: galactose

RBC ionic cloud diagram: Antibodies (IgG and IgM) relevance to hemagglutination

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Harmening, DM: Modern Blood Banking and Transfusion Practices, 5th ed.. F.A. Davis Company, Philadelphia, PA, 2005, Fig. 3-15.



- **DAT and IAT tests detect antibodies** that do not cause direct agglutination, particularly **IgG**, which is too small to hemagglutinate cells itself after centrifugation

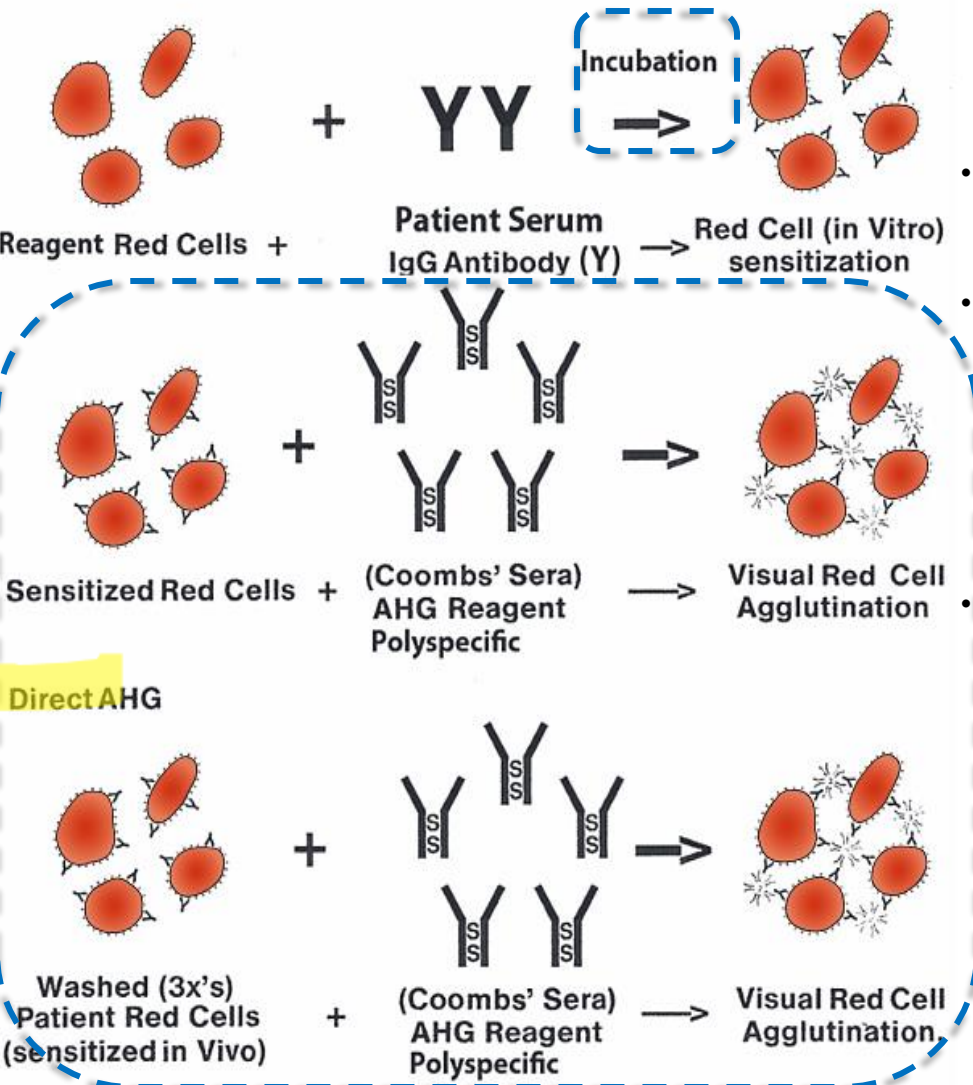
Direct antiglobulin test (**DAT**) detects **IgG/C3** attached to **patient RBCs**, requires **antihuman globulin reagent (AHG, Coombs reagent)** that acts as a bridge to **sensitized cells**

Indirect antiglobulin test (**IAT**) requires **incubating patient's IgG antibodies (serum)** with **reagent cells**

The Antihuman Globulin Test (AHG)/Coombs test indications:

Harmening, DM: Modern Blood Banking and Transfusion Practices, 6th ed.. F.A. Davis Company, Philadelphia, PA, 2012, Fig. 5-3

Indirect AHG



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- **DAT** is part of clinical work-ups for HTR and HDFN
 - Detects **in-vivo sensitization of patient's RBCs** with **IgG/C3**
- **IAT** detects **patient's (serum) unexpected antibodies in-vitro** sensitization of reagent RBCs
 - Part of clinical work-ups for HTR & HDFN
 - **Antibody screen:** Part of **type & screen (T&S)**
 - Determines blood type and screens for **patient unexpected antibodies** against reagent screening cells
- Major **crossmatch** tests donor red blood cells against recipient serum: Necessary for all transfusions
 - Immediate Spin: Detects **ABO incompatibility only**
 - **Incubation** (37°C) for significant unexpected IgG antibodies & **antiglobulin** phases after incubation
 - Type & cross (T&C) involves typing & major crossmatch
 - Removes requested blood from inventory/**unavailable** for others

Rh blood group system

- Clinically, the **most important** blood group **after ABO**
 - Most frequent target of **alloantibodies** caused by transfusions, and in mothers alloimmunized to fetal antigen (**HDFN**)
- The first Rh antigen defined was the D antigen: The term **Rh positive/negative refers to the D phenotype** (A, B, AB, and O, with **+ or – denoting Rh(D)** status)
- But Rh blood group system is comprised of **> 40** individual antigens
 - Most **common Rh antigens** routinely identified in tests are **D (most immunogenic) > c > E > C > e**

Most common transfusion reactions: Estimated frequency

- Transfusion reaction **~1% overall incidence**, most common:
 - **Allergic** (urticarial) 1-3%
 - **Febrile** Non-hemolytic 1-3%
 - **TACO**, Transfusion-Associated Circulatory Overload 1/200
 - **Delayed** Hemolytic Reactions (>24 hours, extravascular) 1/2,500
 - **Septic** transfusion reaction, (**Bacterial** contamination) 1/5000
 - **TRALI**, Transfusion-Related Acute Lung Injury 1/10,000
 - **Anaphylactic** 1/50,000
 - **Acute/Immediate** Hemolytic Reactions (<24 hours, intravascular) 1/75,000
 - Transfusion-associated graft-versus-host disease (**TA-GVHD**) 1/500,000
- Some **overlapping signs and symptoms** → this may lead to misdiagnoses

DHTR (extravascular) and AHTR (intravascular) clinical features: Lab evaluation and management

- **DHTRs** often involve **Rh** system
 - ↓**Hgb**, ↑**bilirubin**, new **antibody**
 - Characterized by **hematocrit** that fails to rise, or decreases
 - Development of a positive **DAT**
- Most common cause of **AHTR** is **ABO** incompatibility
 - Rarely, **Kidd**, **Rh**, **Kell**, and **Duffy** antigen incompatibility also been implicated

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- **AHTR** → **Stop blood transfusion**
- Maintain blood **pressure** (intravenous fluid, vasopressors, diuretics) & **airway**
- **Return** blood and transfusion set to lab & order appropriate labs: Check for serum hemolysis
 - Peripheral blood **schistocytes** & **brown serum**
 - Increased **LDH** (due to RBC lysis)
 - Increased total & indirect **bilirubin** (**jaundice**)
 - Decreased **haptoglobin** (overwhelmed)
 - **Hemoglobinuria** (free hgb in the urine, pink urine)
 - **Hemoglobinemia** (free hgb in the blood)
- Monitor renal function (**BUN**, creatinine)
- Repeat compatibility testing (**crossmatch**)
- Repeat for **DAT**, **IAT**

Delayed Hemolytic Transfusion Reaction

- Onset of symptoms: 5-10 days after RBC transfusion
- S/S: hemolytic anemia, jaundice, fever (can also be asymptomatic)
- Life-threatening complications are rare
- Confirmation: repeat type and screen to detect alloantibody
- Treatment: supportive

Acute Hemolytic Transfusion Reaction

- Abrupt onset of S/S
- S/S: intravascular hemolysis, hypotension, fevers, AKI, pain at the infusion site, DIC, pink plasma or urine
- Treatment: stop the transfusion!
 - Send blood back to blood bank to check for incompatibility, hemolysis
 - Supportive treatment with IVF, pressors, diuresis

Febrile nonhemolytic transfusion reactions

- **FNTR** reactions are due to
 - **Cytokines** released during component storage
 - **Antibodies** in patient's plasma to **donor leukocytes** or **platelets**
- More common in **multiply transfused** patients or **multiparous** women
- **Fever**, often accompanied by **chills**, and sometimes **nausea/vomiting without** evidence of **hemolysis or allergic reaction**
- Reactions usually **more uncomfortable than** clinically **serious**
- Stop and monitor for progression of symptoms, **antipyretics**

Allergic transfusion reactions

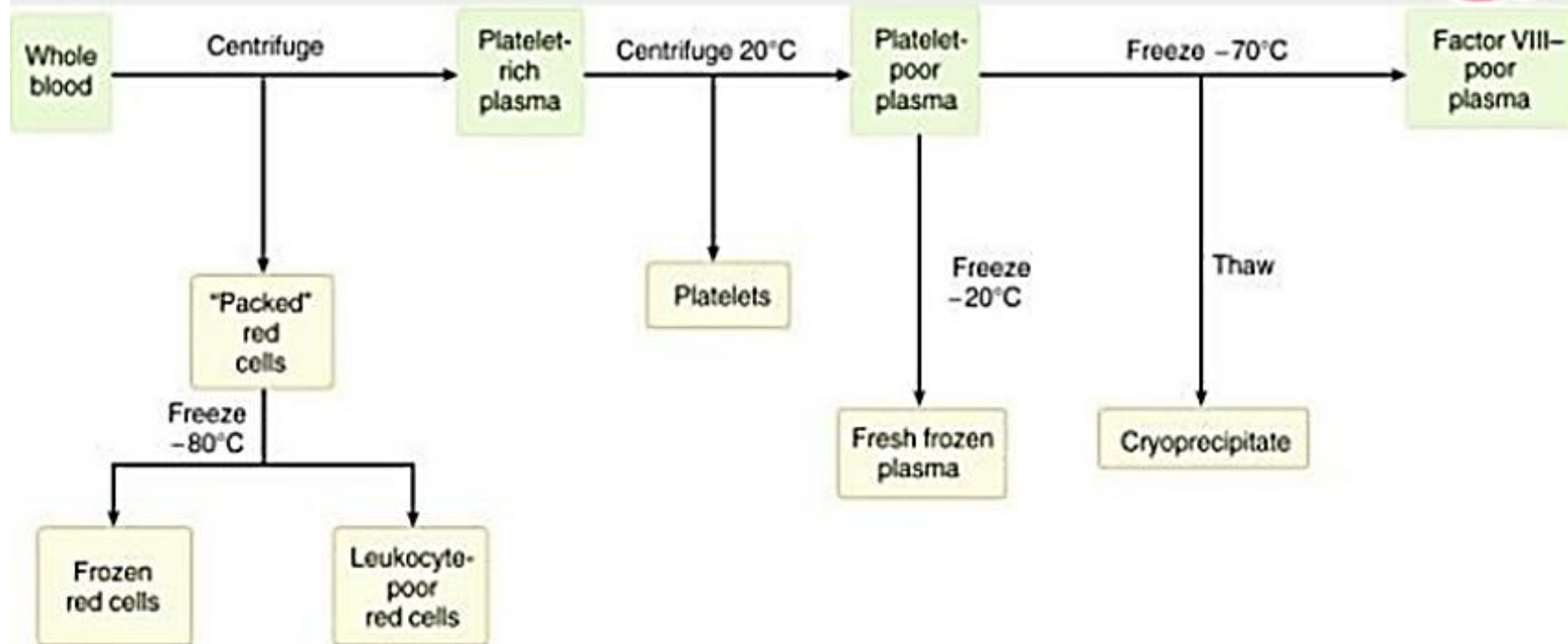
- Caused by **antibodies** from prior exposure to **donor plasma proteins**
- Most common with **plasma-rich components** like platelets or FFP
- Allergic **manifestations** are **immediate**: **Urticaria** common, **bronchospasm**, **anaphylaxis**
 - **Difficulty breathing** symptoms of anaphylaxis require rapid treatment
- Treatment: **Stop transfusion** and **monitor** for **progression**, **diphenhydramine**, possibly **oxygen**, **hydrocortisone** or **epinephrine** for anaphylaxis

Estimated frequency of TTD

- Donor **Screening** Tests

- Hepatitis B: HBsAg, Anti-HBc, HBV DNA
- Hepatitis C: Anti-HCV, HCV DNA
- HIV: Anti-HIV 1/2, HIV-1 RNA
- HTLV: Anti-HTLV I/II
- Syphilis: RPR or IgG to T. pallidum
- West Nile: WNV RNA
- Chagas: IgG to T. cruzi
- Babesia antibodies
- Zika virus RNA
- Hepatitis B <1/800,000 transfusions
- Hepatitis C <1/1,000,000
- HIV <1/1,000,000
- HTLV I and II <1/250,000
- Syphilis No cases >40 yrs
- T.Cruzi ~1:200,000 endemic areas with screening, ~20 cases in US past 20 years
- Babesia <1/100,000 transfusions 2011-2015
- West Nile one per 35 million 2003-2016 donations
- Zika Virus No cases 2016-2019, few cases world-wide

Transfusion therapy principles: Component therapy



Scheme for separation of whole blood for component therapy.



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- **Whole Blood:** 450-500 ml, 60 ml of **citrate** anticoagulant/preservative solution
 - Citrate **toxicity** (**hypocalcemia**)
 - 21-35 days shelf life, **refrigerated**
 - **Uncommon** use today, except trauma/massive transfusions
- **Plasma, platelets** removed as components
 - Packed red blood cells (**pRBCs**) mixed w/ **additive** (ex. Adsol) - (**lower volume** an advantage)
 - Storage period extended (42 days), **refrigerated**

Leukocytes and leukoreduction

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- Leukocytes are **contaminants** of cellular components
- Adverse consequences: **Immunologically-mediated** most **common**), also infectious disease transmission
 - **FNHTRs**, **platelet refractoriness**, transplant **rejection**, **GVHD**, immunosuppression
 - Transmission/reactivation of **viral diseases**
- **Leukoreduction**: Filters (arrow) remove leukocytes
 - Prevents **FNHTRs**, **HLA alloimmunization**, **CMV**



Appropriate RBC transfusion indications

- When **alternative** treatment is **not available**, or **clinical manifestations** in patient demand treatment
 - Levels ≤ 7 gm/dL currently a guideline for **most patients**
 - Patients with **active** coronary artery disease often transfused at higher hemoglobin levels (≤ 8 gm/dL)
- Response to transfusion varies: One red cell unit expected to increase **hemoglobin** by **1 gm/dL** or **hematocrit** by **3%**

Alternatives to transfusions of donor RBCs

- Treat the **underlying cause** of anemia
- **Erythropoietin**
- **Autologous** transfusions for mainly elective surgical procedures: Collection and reinfusion of a patient's own blood
 - Pre-deposit **self-transfusion**
 - Acute normovolemic **hemodilution**
 - **Cell savers**

MTP concept

TABLE

Massive transfusion protocol in obstetrics

	PRBCs	FFP	Platelets	Cryoprecipitate
Round 1	6 U	6 U	6 U	10 U
Round 2	6 U	6 U	6 U	10 U
Round 3	Tranexamic acid 1 g intravenously over 10 min			
Round 4	6 U	6 U	6 U	

Consider activating the protocol when hemorrhage is expected to be massive (anticipated need to replace 50% or more of blood volume within 2 hours), bleeding continues after the transfusion of 4 U of packed red blood cells within a short period of time (1–2 hours), or systolic blood pressure below 90 mm Hg and heart rate is above 120 beats per minute in the presence of uncontrolled bleeding. Once activated, blood bank personnel will continue preparing blood products until the surgical team inactivates the protocol. After round 4, if not inactivated, the protocol will start again from round 1.

FFP, fresh-frozen plasma; PRBC, packed red blood cell; Adapted from Pacheco et al.⁴

Pacheco. Massive transfusion protocols in obstetrics. *Am J Obstet Gynecol* 2016.

- **Activated** by a clinician in **response** to massive/severe bleeding
- **Balanced transfusions** (including pRBCs and clotting factors) used in efforts to **avoid dilutional coagulopathy**

Platelet Transfusion

- Normal platelet **lifespan 5-10 days** stored at **room temperature** with gentle agitation, usually transfused in 5 days
- Indications: **Thrombocytopenia**
 - **<10,000** / μ L to avoid spontaneous hemorrhage
 - **< 50,000** with bleeding or surgery
 - **<100,000** with CNS bleed/surgery
 - Also, bleeding or invasive procedures in patient with platelet **dysfunction**